







IEEE International Conference on Innovation in Computing Data And Electronics (InCODE)

Conference Date: 9th - 10th October 2026

Oriental Institute of Science & Technology, Bhopal

IEEE Record Number: #70499

Paper ID 1451	**LiveLingo: An End-to-End Streaming Middleware for Real-Time Translation of Livestreams into Indic Languages**	Track No. 1

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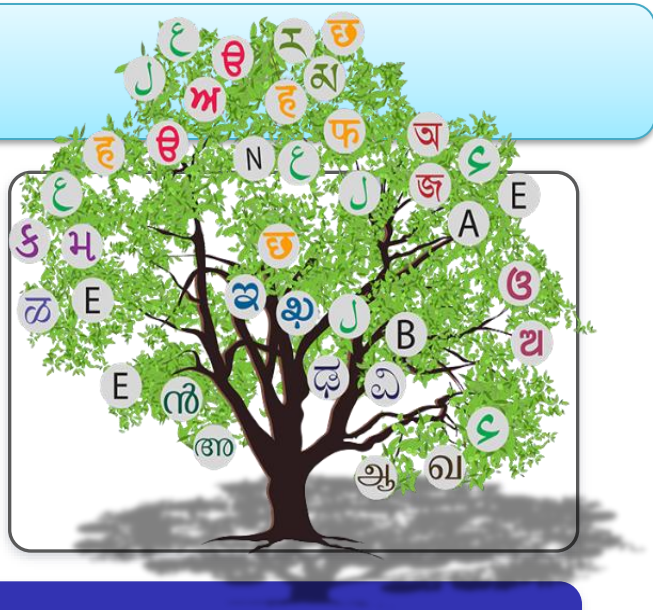
Motivation & Problem Statement

Language barriers

Live video streaming has reshaped global content consumption, yet **language barriers continue to fragment audiences.**

India's diversity

India's linguistic diversity makes **real-time translation** into vernacular languages urgently needed for access.



Cascaded (traditional)

Conventional pipelines cascade **ASR → MT → TTS** — precise, but their composite latency interferes with live-broadcast synchronicity.

ASR → **MT** → **TTS**

Sequential stages → composite latency

vs

End-to-End (LiveLingo)

Multimodal LLMs enable **End-to-End (E2E)** speech-to-speech translation — direct acoustic-to-acoustic mapping that can, in theory, provide near-instantaneous translation.

Direct acoustic → acoustic translation

Single stage → potentially near-instantaneous

LiveLingo connects a continuous live broadcast to Google's **gemini-3.1-flash-live-preview** API over a **persistent WebSocket stream.**

Motivation & Problem Statement

22

languages planned; 18 already interpreted live in Lok Sabha proceedings

19

interpretation booths, 2+ interpreters each, swapped every half hour

5

planned languages not yet operational: Kashmiri, Konkani, Nepali, Sindhi, Santhali

Lok Sabha interpreters relay proceedings in real time in 18 languages

Apart from the original "floor audio", MPs, visitors, and viewers on YouTube can choose from English, Hindi, Assamese, Bengali, Bodo, Dogri, Gujarati, Kannada, Maithili, Malayalam, Manipuri, Marathi, Odia, Punjabi, Sanskrit, Tamil, Telugu or Urdu.

Updated – March 27, 2025 11:10 pm IST – NEW DELHI

AROON DEEP

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A few minutes before parliamentary proceedings start at 11 a.m., an interpretation cell administrator takes a quick roll call of each of the 19 interpretation booths, making sure they are staffed with at least two people, who swap places every half an hour (interpretation is a mentally taxing work, necessitating such swaps). A total of 22 languages are planned — booths for Kashmiri, Konkani, Nepali, Sindhi and Santhali were not, as of Wednesday (March 26, 2025), operational. Urdu and Sanskrit interpretation feeds are not yet being streamed on YouTube.

Source: The Hindu, Mar 2025

Research Gap & Objectives

What is known

Prior E2E systems demonstrate **low-latency translation**

Research gap

Sustained stateful WebSocket streaming under real-world network degradation is **insufficiently characterized**, particularly for **Indic languages**.

Study objectives

1 Latency

Compare **Time-to-First-Audio (TTFA)** latency in the continuous streaming pipeline under **optimal vs. constrained** network conditions.

TTFA: stable vs. constrained

2 Accuracy vs. latency

Statistically characterize the relationship between **semantic translation accuracy** and **transmission latency** — and determine whether it is continuous or threshold-governed.

Continuous or threshold?

3 Error analysis

Conduct an **error analysis** of translated output to understand how multimodal foundation models behave when acoustic input is fractured by network instability.

Behaviour under instability

Related Work

E2E Speech-to-Speech Translation

- SeamlessM4T [1] — direct acoustic mapping, no text intermediation
- Seamless Streaming [2]
- AudioPaLM [4] — adds audio capability to LLMs

Streaming / Simultaneous Translation

- Wait-k chunked encoders [3],[6]
- Typically require local buffering + voice activity detection (VAD) before translation begins

Deployed Real-Time Systems

- LinguaLive [8] — multimodal real-time translation
- Google Meet Live Audio Translation — Gemini Live API, direct audio-to-audio
- **Neither characterises sustained network degradation**

Low-Resource Indic Translation

- Samanantar [9], IndicVoices [10] — parallel corpora
- XLS-R [12], mSLAM [13] — cross-lingual representations
- Fidelity drop typically observed when context is lost [7]

What exists: simulated streaming / fixed datasets → **What's missing:** behaviour under sustained, real network degradation → **What LiveLingo studies:** continuous streams under realistic network jitter

Historical arc

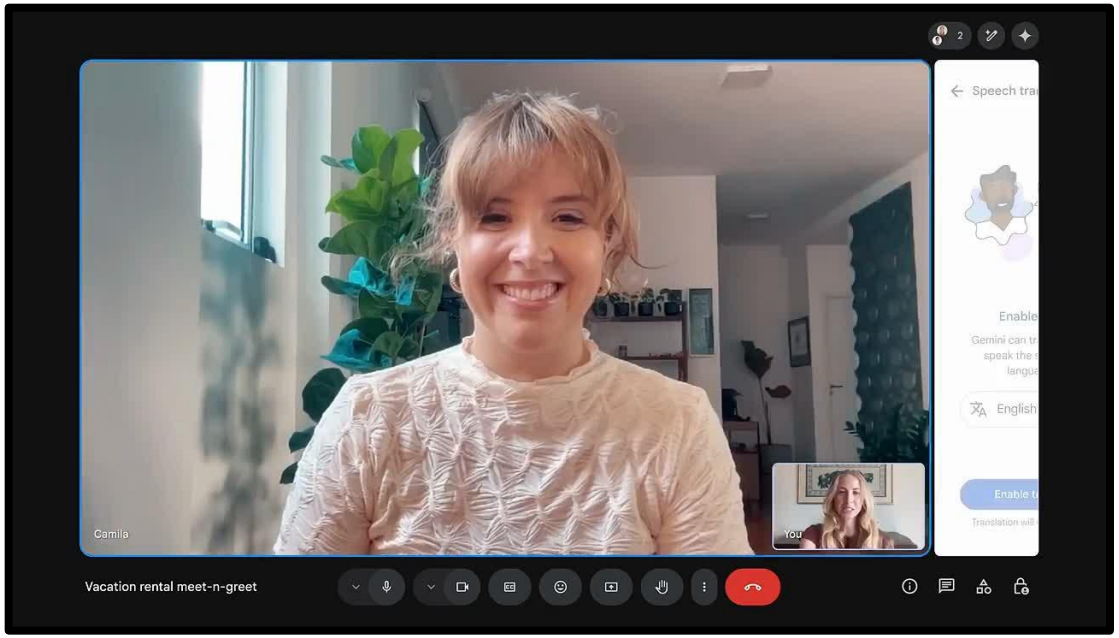
2000
Verbmobil, early cascaded MT (Wahlster)

2019
Direct speech-to-speech models (Jia et al.)

2019
Wait-k simultaneous streaming (Ma et al.)

2024
Multimodal LLMs, e.g. Gemini (Google DeepMind) → this work

Related Work

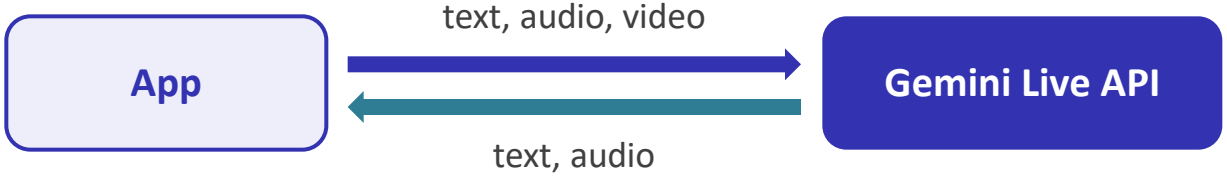


Demo of Google Meet Live Audio Translation (Example of End-to-End Architecture)

Scan to watch

Google Meet Live Audio Translation demo

Deployed precedent: Gemini Live API (used by Google Meet)



Connected over a **persistent WebSocket** — the same pattern LiveLingo studies under network stress.

LiveLingo Architecture

Capture & prepare (local)

1 Live HLS Broadcast
English-language livestream source

2 Ingestion & PCM Extraction
Demux in real-time → 16 kHz mono, 16-bit PCM

3 Speech Segmentation
Local energy-based detector (no ASR/MT stage)

Translate (streaming)

4 Persistent Stateful WebSocket
Full-duplex; connection setup paid per session

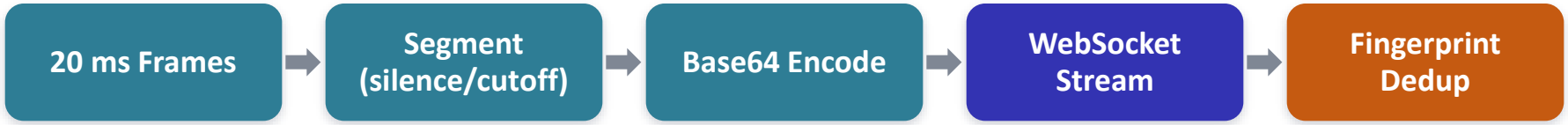
5 Gemini 3.1 Flash Live Preview
Acoustic-to-acoustic translation; audio-only response

Deliver (output)

6 Deduplication / Multiplexing
Time-domain fingerprint filter suppresses looping

7 AWS IVS Output Stream
Translated Hindi / Bangla channels

Methodology: Streaming Pipeline

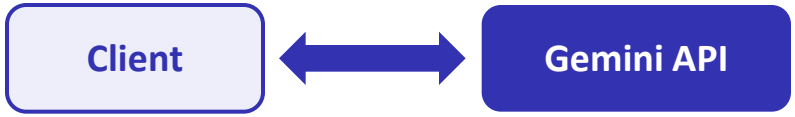


Segmentation

- **20 ms** analysis frames; energy threshold = **0.016**
- **400 ms** silence → segment emitted; **10 s** hard cutoff
- Segments base64-encoded; **no local ASR/MT stage**

Transmission

- Persistent, **full-duplex** bidirectional WebSocket
- Response modality: **audio only** (translated speech)



Deduplication / Multiplexing

- **14-dimensional** fingerprint: 12 amplitude bins + RMS energy + zero-crossing rate
- **2500 ms** sliding comparison window
- Cosine similarity threshold = **0.992** → duplicate dropped before AWS IVS remux

Benchmark Harness

1200 ms

fixed segmentation

E2E and cascaded pipelines receive identical input.

Experimental Setup

Dataset

- **English speech:** public presidential speeches
- **5-minute and 10-minute** passages (10-min is a superset of 5-min)
- 16 kHz mono, 16-bit PCM
- Same source audio for Hindi & Bangla

Languages

Hindi
high-resource

Bangla
low-resource

10 **independent runs per condition**

Fresh session each run

Network conditions

Stable

10 Mbps

0% loss

<5 ms jitter

Constrained

~350 Kbps

5% loss

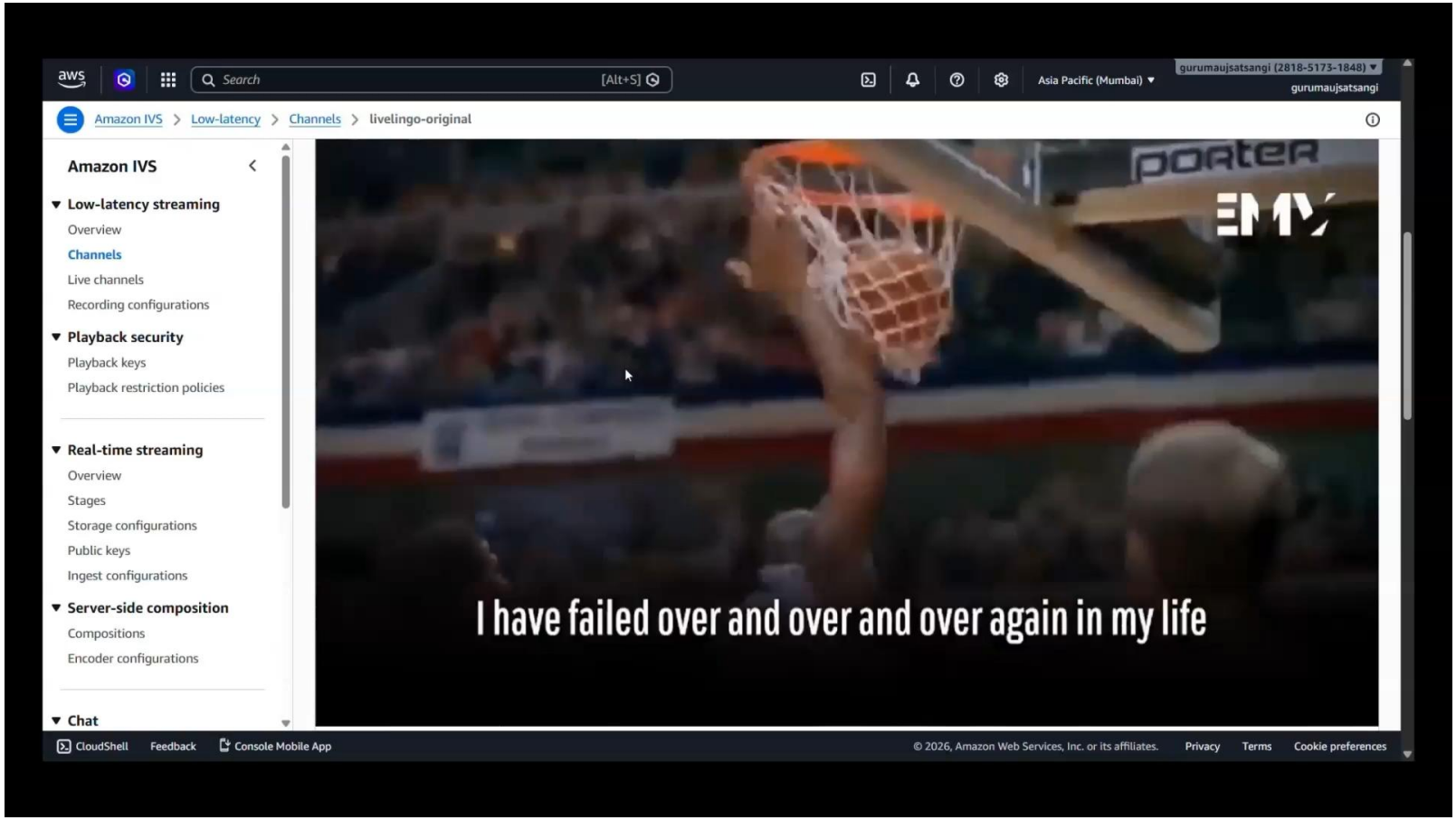
50 ms jitter

Cascaded baseline (Sarvam API)

ASR	saaras:v3
MT	sarvam-translate:v1
TTS	bulbul:v3

Demo Video

LiveLingo in action



Live demo: a continuous English stream translated in real time and republished via AWS IVS

Evaluation Metrics

TTFA

Time to First Audio

Duration (ms) from source-frame ingestion to output-frame generation.

ms latency unit

Semantic Accuracy

LaBSE Cosine Similarity

Cosine similarity between generated translation transcripts and ground-truth references.

0–1 similarity score

MOS

Subjective Audio Quality

Double-blind ACR test, 5-point Likert scale (ITU-T P.800).

1–5 listener rating

MOS rater panel

51 **native speakers**

26 Hindi · 25 Bangla

Statistical significance

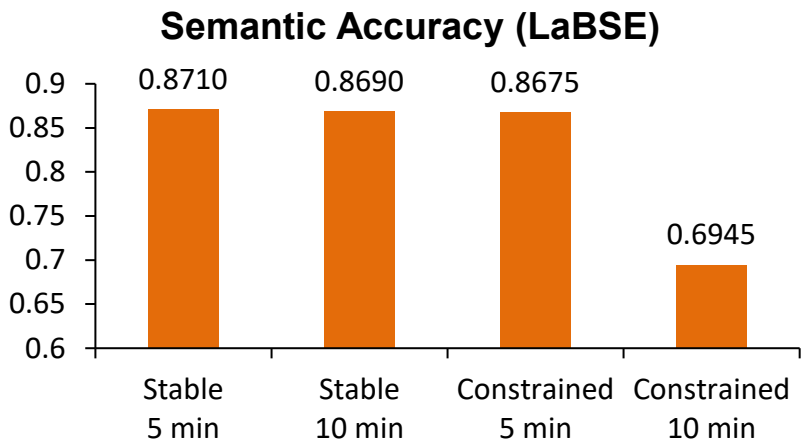
$\alpha = 0.05$

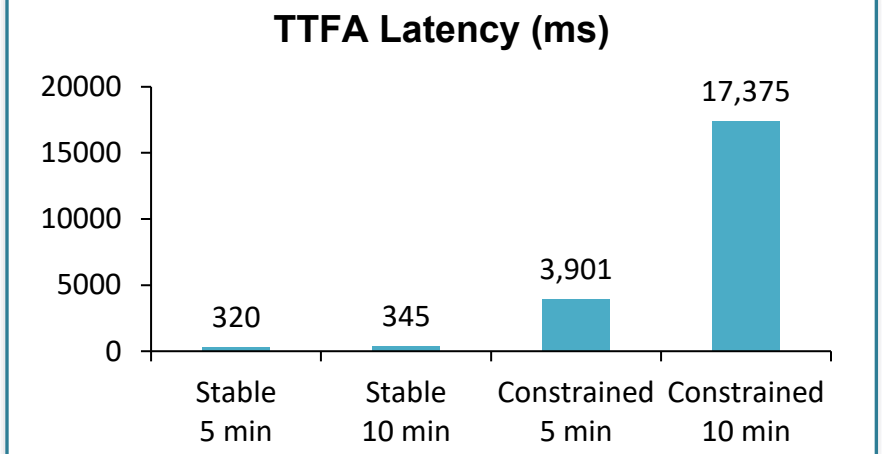
Two-sample Welch's unequal-variance t-test

Results: Stable vs Constrained Network

Network	Duration	TTFA (ms)	LaBSE Accuracy
Stable (10 Mbps)	5 min	320.00 ± 14.20	0.8710 ± 0.012
Stable (10 Mbps)	10 min	345.00 ± 18.50	0.8690 ± 0.015
Constrained (~350 Kbps)	5 min	3900.50 ± 210.40	0.8675 ± 0.018
Constrained (~350 Kbps)	**10 min**	**17375.00 ± 412.50**	**0.6945 ± 0.031**

! **Constrained 10-min:** TCP head-of-line blocking → latency spikes to **17,375 ms**; accuracy falls to **0.6945**.





Threshold Effect: Acoustic Context Fragmentation

Pooled

r = −0.926

p = 0.008

significant

Excluding extreme

r = −0.588

p = 0.297

not significant

Failure is a discontinuous threshold effect, not progressive degradation.

Below threshold

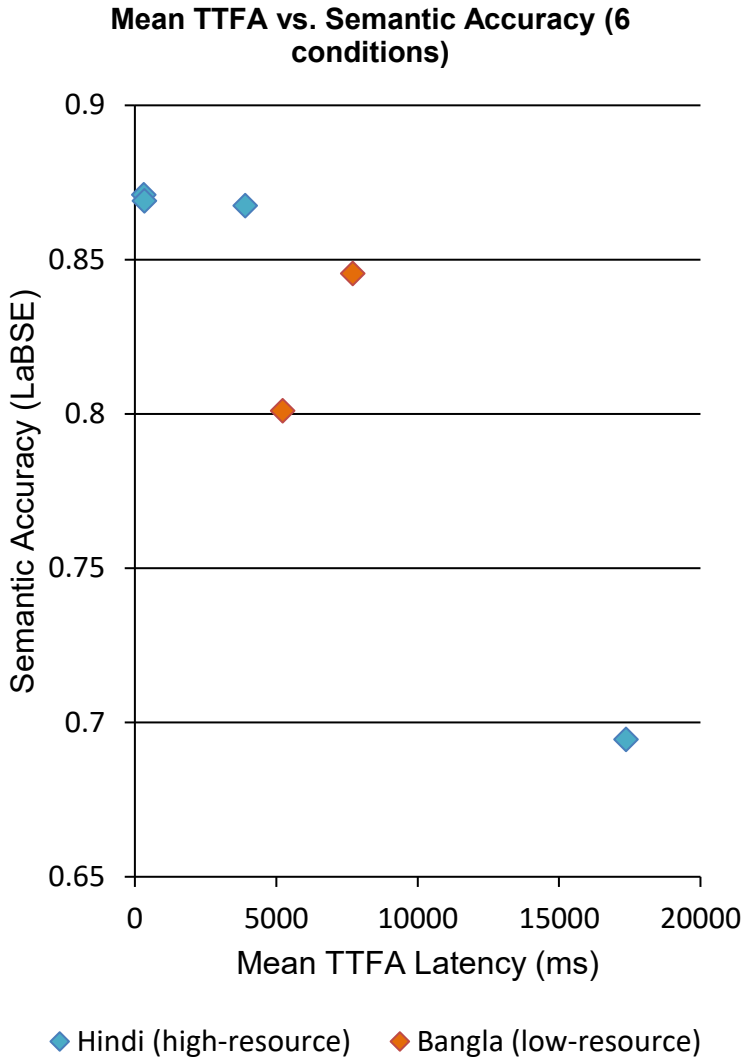
- Queueing delay only
- Acoustic context stays contiguous
- Accuracy largely preserved

Above threshold

- Acoustic Context Fragmentation
- Decoder gets incomplete context
- Semantic drift / hallucination

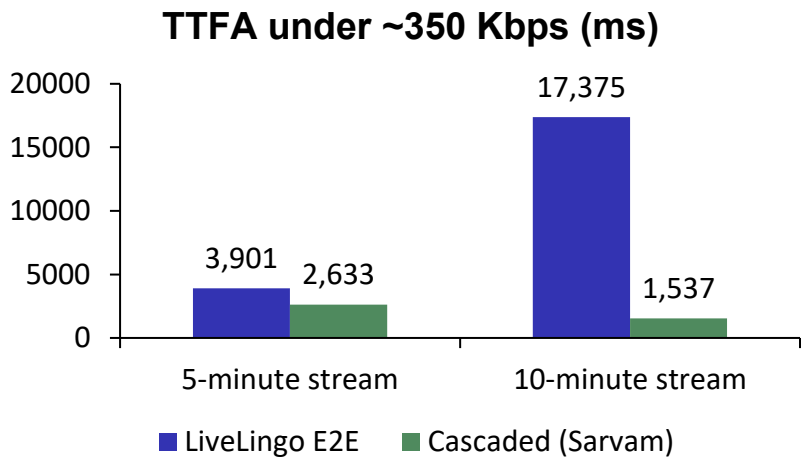
Threshold bracketed between **7.70 s and 17.38 s**

Observed bracket for this recording — not an established universal architectural constant.



E2E vs Cascaded Architecture (Constrained Network)

Metric	LiveLingo (E2E)	Cascaded (Sarvam)
5-min TTFA	**3,900.50 ms**	**2,633.34 ms**
10-min TTFA	**17,375.00 ms**	**1,537.00 ms**
10-min Accuracy (LaBSE)	**0.6945**	**0.8690**



E2E: Persistent Stateful WebSocket

- Potentially ultra-low latency
- Sensitive to severe network stalls

Cascaded: Stateless Per-Segment

- Higher baseline latency (ASR→MT→TTS)
- More resilient to network jitter/degradation

E2E can provide sub-second latency under stable conditions, while the cascaded baseline showed **greater resilience under the constrained condition** tested.

Hindi vs Bangla: Accuracy vs. Naturalness

Language	10-min TTFA (constrained)	Semantic Accuracy	MOS (raters n)
Hindi (high-resource)	17375.00 ± 412.50 ms	0.6945 ± 0.031	4.73 ± 0.22 (n=26)
Bangla (low-resource)	7703.50 ± 310.20 ms	0.8455 ± 0.021	4.53 ± 0.28 (n=25)

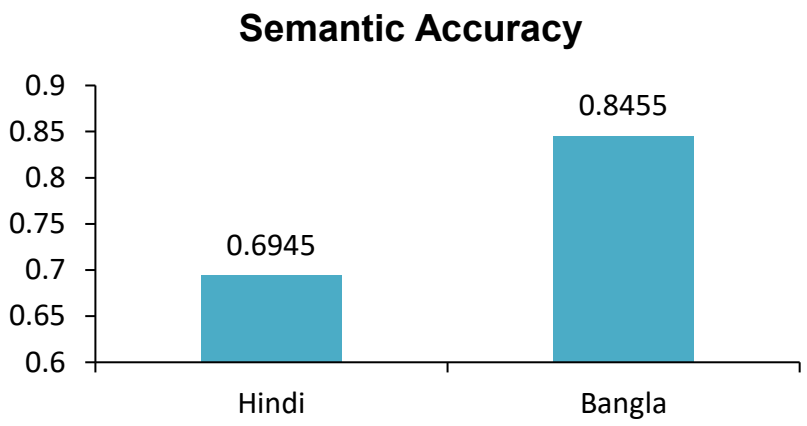
0.6945
Hindi semantic accuracy

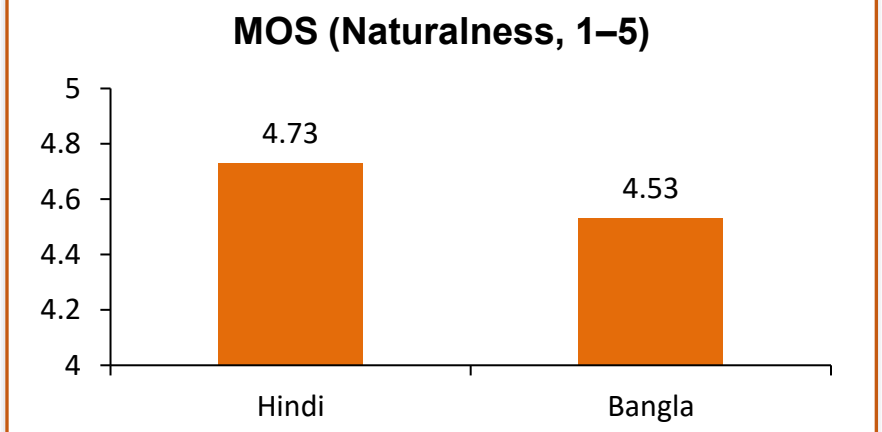
≠

4.73
Hindi MOS (naturalness)

Key observation: semantic correctness and acoustic naturalness can diverge.

Even when Hindi's accuracy degraded substantially, human listeners still rated the audio as highly natural.





Conclusion & Novelty

1 Fast when stable

E2E streaming achieves **sub-400 ms TTFA** under stable 10 Mbps conditions.

2 Fragile under constraint

Severe network constraints expose a **major weakness** of stateful WebSocket streaming.

3 Threshold, not gradual

Accuracy degradation is better characterized as a **threshold effect** than a continuous latency effect.

4 Root cause

Acoustic Context Fragmentation explains the observed catastrophic semantic degradation.

Novelty / Contribution

Empirical characterization of continuous, stateful E2E streaming under sustained network degradation — including the observed **threshold behaviour** and the **Acoustic Context Fragmentation** phenomenon.

Limitations & Future Work

Limitations

- Synthetic network throttling, not real-world mobile networks
- N=51 cohort from a single university
- Single-source speech corpus
- Only Hindi and Bangla evaluated
- Limited generalizability across speakers, dialects, language families
- **Acoustic interference:** noise/overlapping speakers can trigger VAD inaccuracies
- **Dialectal nuance:** regional idioms/slang/jargon may reduce semantic precision

Future Work

1. Controlled latency sweep to localize the threshold precisely
2. Adaptive streaming / 500 ms micro-batching; adaptive frame pacing
3. Larger, multi-speaker datasets; more Indic languages
4. Context-preserving cross-lingual representations

Thank You for your attention!

Questions?